

# LIOR SHAHVERDI

Senior Software Engineer | Healthcare AI, LLMops, Clinical Evaluation Systems

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## PROFESSIONAL SUMMARY

Senior Software Engineer with deep experience building production healthcare AI systems, LLM evaluation infrastructure, and scalable clinical data platforms. Strong fit for senior AI/backend roles requiring reliable LLM workflows, safety-oriented model routing, ground-truth evaluation, EHR data pipelines, and cross-functional ownership in high-stakes healthcare environments.

## CORE SKILLS

|                        |                                                                                                                                                                    |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Healthcare AI / LLMops | Azure Prompt Flow, MLflow, multi-step clinical AI workflows, prompt/model versioning, model comparison, safety-oriented routing, clinical decision support, MCP    |
| Evaluation / Safety    | ground-truth datasets, failure/mismatch analysis, alignment metrics, uncertainty handling, abstention/escalation patterns, longitudinal behavior monitoring        |
| Data / ML Platforms    | Databricks, Unity Catalog, Airflow, Spark/PySpark, Delta Lake, EHR transaction ingestion, experiment logging, drift analysis, vector search                        |
| Backend / Cloud        | Python, Scala, Java, JavaScript, FastAPI, Flask, Django, REST, GraphQL, RabbitMQ, Postgres, Redis, Cassandra; Azure AI Foundry, Azure SDK, AWS EC2/S3/RDS/DynamoDB |

## SELECTED PROJECTS

- DevMemoryIndex** — [github.com/liorshahverdi/devmemoryindex](https://github.com/liorshahverdi/devmemoryindex) — Built a persistent developer memory layer that captures workflow knowledge from git commits, terminal history, markdown notes, meetings, browser activity, and AI-agent sessions. Implements LanceDB-backed hybrid semantic/keyword retrieval, rich metadata, context-building for LLM prompts, a 21-command Typer CLI, FastAPI REST API, background daemon, git hooks, and 19 Claude Code MCP tools for agent memory, planning, scoring, graph traversal, and store-health analysis.
- Markdown Notes** — [LINK PLACEHOLDER] — Built a local-first markdown knowledge system with vault browsing, source ingestion, Mermaid-aware editing, semantic search, and RAG-powered chat over a document corpus using SvelteKit, TypeScript, Ollama, and local embeddings; designed around private/offline knowledge retrieval rather than cloud-hosted note storage.
- voice-news** — [LINK PLACEHOLDER] — Built a privacy-first news-to-audio pipeline that concurrently aggregates RSS, Hacker News, and Reddit, deduplicates overlapping stories, generates podcast-style scripts with a local LLM, and synthesizes speech with on-device TTS, creating an end-to-end offline AI content workflow.
- PhysIQ** — [LINK PLACEHOLDER] — Built an offline-first fitness coaching system with Rust TUI and Kotlin Android clients, voice-input meal/exercise tracking, local LLM recommendations, GPS activity logging, and LAN-based device sync; emphasizes private health-adjacent data handling and local AI inference.

## PROFESSIONAL EXPERIENCE

- Johns Hopkins Medicine — Senior Machine Learning Engineer** Dec 2022 – Present
- Designed end-to-end ML/LLM evaluation frameworks for clinical decision-support systems across antimicrobial stewardship and VTE workflows, spanning data ingestion, experimentation, metric definition, evaluation, and stakeholder reporting.
  - Built ground-truth generation and schema-standardization pipelines that transform annotated clinical outputs into reusable experimentation datasets for LLM benchmarking, prompt iteration, and deterministic indicator alignment.
  - Developed multi-step clinical AI experimentation pipelines using Azure Prompt Flow and MLflow, including prompt-driven evaluation, decision-tree routing logic, model/prompt versioning, cost-aware execution, and repeatable failure analysis.
  - Led integration and evaluation of John Snow Labs annotation workflows, modeling annotation lifecycle states, extracting normalized ground truth, and balancing manual labeling, pre-annotation, and LLM-assisted labeling strategies.
  - Own evaluation strategy for healthcare AI systems: labeling fidelity, citation quality, mismatch analysis, uncertainty handling, abstention/escalation patterns, and longitudinal monitoring of model behavior and drift.
  - Architect Databricks, Unity Catalog, Spark/PySpark, Delta Lake, Azure SQL, blob storage, and Airflow pipelines for EHR ingestion, clinical feature extraction, downstream logging validation, and LLM/ML analytics.
  - Serve as technical bridge across ML, data engineering, clinical, and compliance stakeholders; participate in SIP reviews, code reviews, architecture review, and prioritization for internally developed AI systems.
  - Selected to lead radiology de-identification initiative, owning technical direction and implementation planning for privacy-sensitive clinical AI/data workflows.
- Critical Mention — Team Lead** Apr 2019 – Dec 2022
- Promoted to Team Lead to define architecture, manage release engineering, and mentor distributed backend/infrastructure engineers.
  - Designed fault-tolerant Python/RabbitMQ microservices and Flask REST APIs powering low-latency real-time search and high-throughput partner data ingestion.
  - Implemented ML pipelines for entity deduplication and metadata extraction across large-scale unstructured text datasets, improving data quality for search and analytics workflows.
  - Engineered resilient streaming pipelines consuming third-party SDKs and high-velocity partner APIs for real-time audience analytics; automated AWS SDK provisioning to reduce release friction.
- TechnoFist.fm — Web Developer** Nov 2017 – Jan 2020
- Built a proprietary CMS optimized for media tracking, editorial workflows, and community-driven content organization.
- Violerts.com — Software Engineer** Mar 2018 – Jun 2018
- Developed JavaScript ingestion interfaces for real-time city-agency feeds and ArcGIS-based spatial data visualization.
- NYC Department of Buildings — Data Analyst** Dec 2015 – Apr 2017
- Modeled operational metrics from Oracle data warehouses to build optimized executive dashboard infrastructure.

## EDUCATION

The College at Brockport, State University of New York — Bachelor of Science (B.S.) in Computer Science